Indroneel Roy

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Education

Shahjalal University of Science and Technology (SUST) February 2020 – August 2025
B.Sc. in Electrical & Electronic Engineering

Research Projects

EdgeMambaFormer: Edge-Guided Cross-Scale Mamba Transformer with Wavelet-Based Boundary Attention for Colonoscopy Polyp Segmentation | *PyTorch, Mamba/SSM* GitHub ↗

- Proposed EdgeMambaFormer, a novel tri-hybrid architecture integrating a PVT v2-B2 CNN-Transformer encoder, a Wavelet Edge Attention Gate (WaveletEAG), a Cross-Scale Mamba Module (CSMM), and a Dual-Branch Transformer Decoder for colonoscopy polyp segmentation.
- Designed WaveletEAG, a mathematically grounded, parameter-free boundary attention module using a 2-D Haar Discrete Wavelet Transform to extract multi-directional edge signals, fused with high-level semantic context via a learnable sigmoid gate.
- Proposed CSMM, a bidirectional selective-scan State Space Model (Mamba) that jointly tokenizes multi-scale encoder features into a unified sequence, enabling cross-scale global context modeling with linear complexity.
- Achieved best-in-class **mDice/mIoU** on 3 of 4 independently trained colonoscopy benchmarks (Kvasir-SEG: **93.5%**/88.4%; CVC-ClinicDB: **94.1%**/89.2%; CVC-ColonDB: **88.9%**/81.2%), with the largest margin on CVC-ColonDB (+8.1% mDice over Polyp-PVT), using a compact 25.43M-parameter model.

MSHFNet: Multi-Scale Hybrid Fusion Network for Multi-Organ CT Segmentation | *PyTorch, timm* GitHub ↗

- Proposed MSHFNet, a dual-pathway encoder-decoder architecture integrating a pretrained ResNet-50 CNN encoder with a Swin-Base Transformer encoder, designed for precise multi-organ segmentation on abdominal CT scans.
- Designed a novel Cross-Attention Fusion Module (CAFM) performing bidirectional cross-attention at four hierarchical scales, enabling each modality to selectively enrich the other beyond simple concatenation or addition.
- Implemented a Shared Decoder with Deep Supervision using a combined Cross-Entropy, Dice, and Boundary loss, providing auxiliary gradient signals at three intermediate resolutions to improve boundary-aware learning.
- Achieved **79.73%** mean DSC on the Synapse multi-organ benchmark (8 organs), surpassing TransUNet by +2.25% DSC, with state-of-the-art Pancreas segmentation at **81.47%**; further achieved **91.18%** mean DSC on the ACDC cardiac MRI benchmark, confirming cross-domain generalisability.

Additional Projects

PolyFormer: Hybrid CNN-Transformer Network for Polyp Segmentation | *PyTorch, OpenCV* GitHub ↗

- Designed a novel encoder-decoder architecture integrating a pretrained **ResNet-50 CNN** backbone with a custom **Transformer bottleneck** featuring learnable positional encodings and 8-head multi-head self-attention for global context modeling in colonoscopy images.
- Developed a **Cross-Attention Fusion Module** where Transformer features attend to CNN spatial features as key-value pairs, enabling explicit synergy between local texture representations and long-range spatial dependencies.
- Implemented a **Combined Dice + BCE loss** with a U-Net style hierarchical decoder using four skip-connection stages, improving boundary precision for small and irregularly shaped polyps.
- Achieved **Dice: 0.8971** and **IoU: 0.8172** on the **Kvasir-SEG** benchmark (60 epochs, NVIDIA T4 GPU), trained with AdamW optimizer and Cosine Annealing scheduler.

Cassava Leaf Disease Classification using Vision Transformer (ViT) | *PyTorch, timm* GitHub ↗

- Designed and fine-tuned a ViT-Base/16 model for multi-class plant disease classification across 5 categories on the Kaggle Cassava Leaf Disease dataset (21,367 training images).
- Achieved **87.15%** validation accuracy with **93.4%** precision and **95.6%** recall for Cassava Mosaic Disease (CMD), outperforming ResNet-50 baseline by ∼4.2%.
- Implemented full training pipeline with transfer learning via timm, GPU-accelerated 224×224 preprocessing, label smoothing, and cosine LR scheduling.

Vision-Based Wheat Head Detection using YOLOv11 | *PyTorch, OpenCV* GitHub ↗

- Fine-tuned YOLOv11-Medium on a large-scale wheat head detection dataset with **29,422+** annotated instances for precision agriculture applications.
- Achieved **94.7% mAP@50** and **55.4% mAP@50–95** with **91.7%** precision and **89.2%** recall, demonstrating strong generalization under varying occlusion and lighting.

Agentflow: Multi-Tool Conversational AI System | *LangChain, LangGraph, LLM* GitHub ↗

- Designed a stateful conversational AI system using LangGraph state machine architecture with persistent SQLite checkpointing for robust multi-turn dialogue management.
- Integrated 10+ specialized tools including web search, financial APIs (Alpha Vantage, CoinGecko), and weather services with dynamic tool binding using Groq LLM (Llama-3.3-70B).

Model Implementations

- **CNN Architectures** — Implemented and trained LeNet, AlexNet, ResNet, DenseNet, VGG16 and EfficientNet from scratch in PyTorch. GitHub ↗
- **Transformer** — Implemented the Transformer architecture from scratch, including self-attention, multi-head attention, positional encoding, layer normalization, encoder, and decoder modules. GitHub ↗
- **Image Segmentation Models** — Implemented FCN, Mask R-CNN, U-Net (base, +CBAM, +SE), Non-Local U-Net, and TransUNet; explored hybrid CNN-Transformer fusion strategies (dual-path, serial, bottleneck, and skip-connection attention) alongside pure Transformer encoders (SETR) and Swin-UNet for medical and natural image segmentation. GitHub ↗
- **Diffusion Models** — Implemented Denoising Diffusion Probabilistic Models (DDPM), Denoising Diffusion Implicit Models (DDIM), Classifier-Free Guidance (CFG), and Latent Diffusion Models (LDM). GitHub ↗
- **Semi-Supervised Learning** — Implemented Mean Teacher, and graph-based/transductive and wrapper/inductive semi-supervised learning methods; investigated pseudo-labeling and consistency regularization techniques for leveraging unlabeled data. GitHub ↗

Technical Skills

Languages: Python, SQL, C, MATLAB, Assembly (Intel ×86)

ML/DL Frameworks: PyTorch, TensorFlow, Keras, scikit-learn, OpenCV

Computer Vision: CNNs, Object Detection (YOLO), Image Segmentation, ResNet, U-Net, ViT, Swin Transformer, Mamba/SSM, Diffusion Models (DDPM/DDIM)

Semi-Supervised Learning: Mean Teacher, Pseudo-Labeling, Consistency Regularization, Transductive/Inductive Methods

LLM & NLP: LangChain, LangGraph, Transformers, Hugging Face, RAG, Vector Databases

MLOps & Deployment: Docker, FastAPI, Streamlit, Hugging Face Spaces, Git, AWS

Databases: SQL, Pinecone, ChromaDB

Algorithms & DS: Data Structures & Algorithms 200+ LeetCode problems solved

Certifications

Neural Networks and Deep Learning — Coursera / deeplearning.ai (View Certificate)

Additional Information

Languages: English (Fluent), Bengali (Native)

Interests: Open-source AI contributions, research paper implementations, technical blogging, chess, cooking, table tennis, badminton